

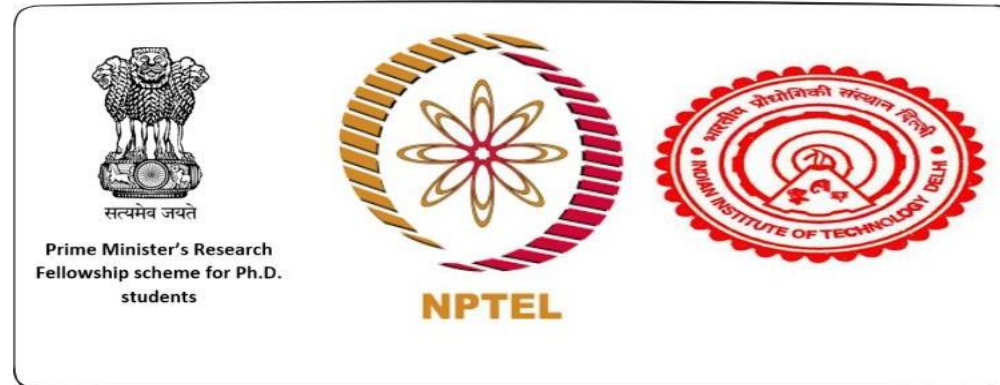
Interaction Session 4

*on*

***Cloud Computing***  
***noc25-cs11***

*by*

**Mainak Chakraborty**



21<sup>st</sup> February 2025  
Indian Institute of Technology, Delhi

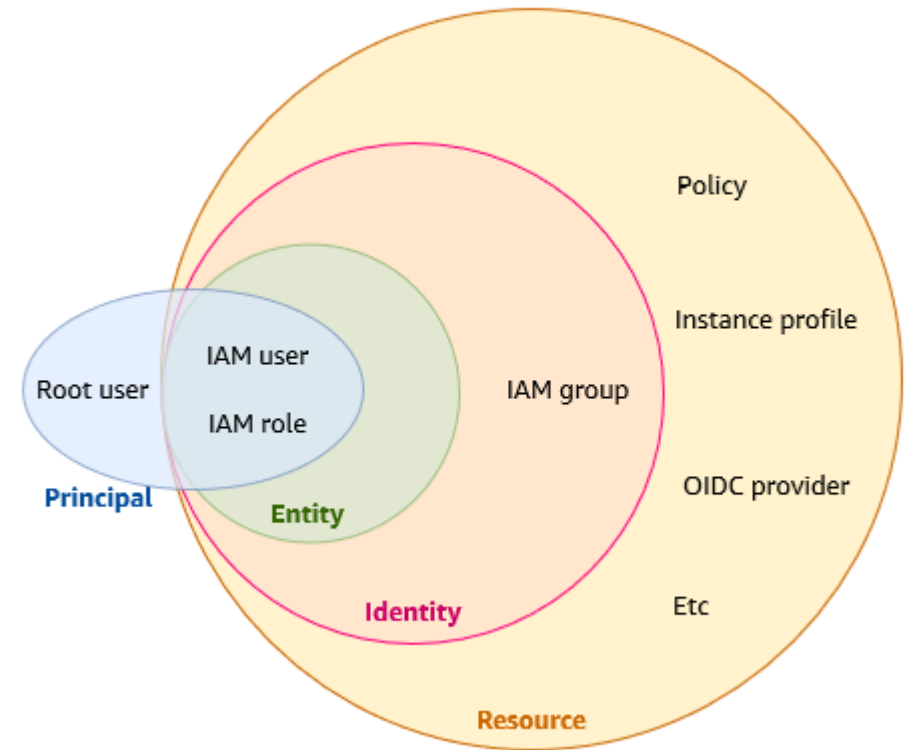
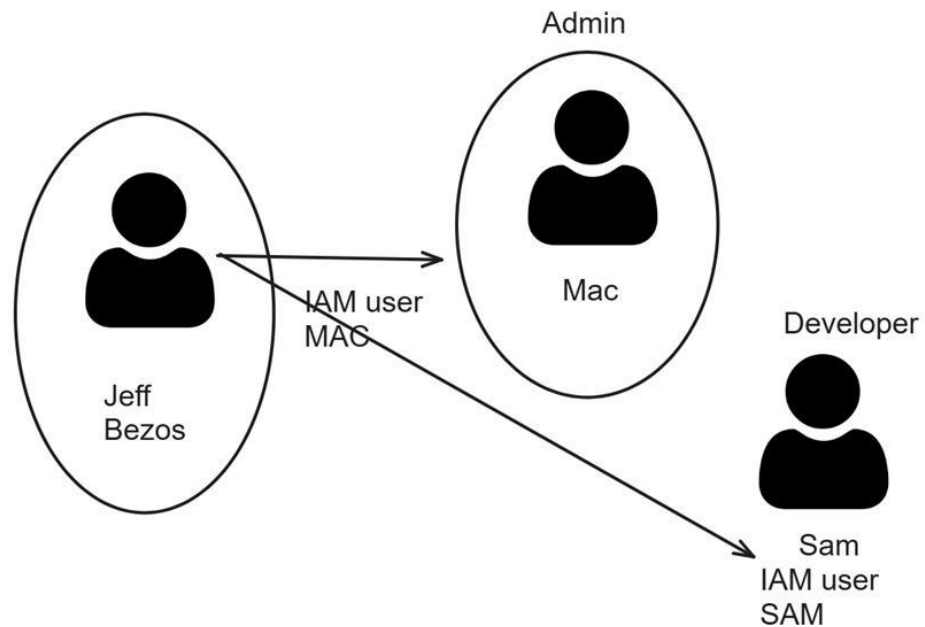
## Points to Remember:

- Total 8 more Discussion sessions to be taught lectures and Tutorials
- Any specific problem, please upload in the discussion forum, I will try to reply within 24 hours
- PPT and Video will be shared with everyone
- Assignments will not be solved but similar problems will be solved.

# TOPICS TO BE COVERED

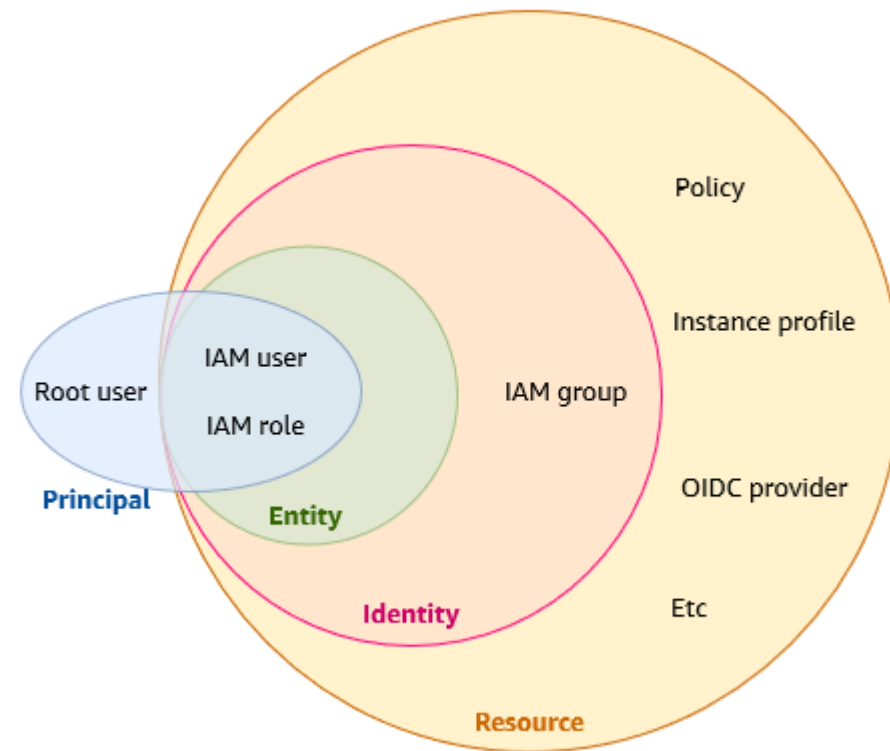
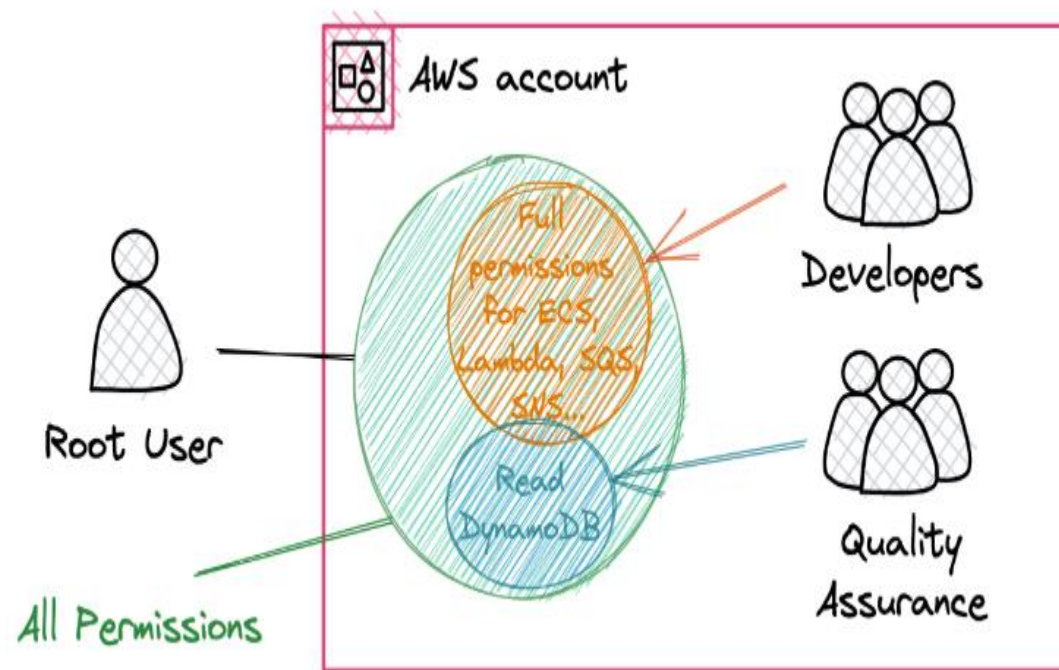
1. AWS-Hands on session
2. Root User vs IAM users
3. Create IAM user
4. Multi-Factor Authentication
5. Amazon EC2

# Background



Source: [https://docs.aws.amazon.com/IAM/latest/UserGuide/introduction\\_identity-management.html](https://docs.aws.amazon.com/IAM/latest/UserGuide/introduction_identity-management.html)

# IAM User



Source: <https://blog.awsfundamentals.com/aws-iam-users>

# Root User vs IAM user

Resources, while an IAM user has custom permissions.

|                    | Root user                                               | IAM user                                            |
|--------------------|---------------------------------------------------------|-----------------------------------------------------|
| Access             | Unrestricted access to all account resources            | Custom permissions set by the account owner         |
| Creation           | The first identity created when you create your account | Can be created by the root user or another IAM user |
| Number per account | One per account                                         | Multiple users can be created                       |
| Permissions        | Can't be restricted                                     | Can be limited by policies                          |
| Use cases          | Create your first IAM user                              | Represent a person or application                   |

# Json

## JSON vs XML

- JSON is lightweight thus simple to read and write.
- JSON supports array data structure.
- JSON files are more human readable.
- JSON has no display capabilities .
- Provides scalar data types and the ability to express structured data through arrays and objects.
- Native object support.
- XML is less simple than JSON.
- XML doesn't support array data structure.
- XML files are less human readable.
- XML provides the capability to display data because it is a markup language.
- Does not provide any notion of data types. One must rely on [XML Schema](#) for adding type information.
- Objects have to be expressed by conventions, often through a mixed use of attributes and elements.

### Similarities between JSON and XML

Both are simple and open.

Both supports unicode. So internationalization is supported by JSON and XML both.

Both represents self describing data.

Both are interoperable or language-independent.

```
<?xml version="1.0" encoding="UTF-8"?>
<DatabaseInventory>
  <DatabaseName>

  <GlobalDatabaseName>production.cubicrace.com</GlobalDatabaseName>
  <OracleSID>production</OracleSID>
  <Administrator EmailAlias="piyush"
  Extension="6007">Piyush
  Chordia</Administrator>
  <DatabaseAttributes Type="Production"
  Version="9i" />
  <Comments>All new accounts need to be
  approved.</Comments>
</DatabaseName>
<DatabaseName>

  <GlobalDatabaseName>development.cubicrace.com</GlobalDatabaseName>
  <OracleSID>development</OracleSID>
  <Administrator EmailAlias="kalpana"
  Extension="6008">Kalpana
  Pagariya</Administrator>
  <DatabaseAttributes Type="Development"
  Version="9i" />
</DatabaseName>
</DatabaseInventory>
```

```
{
  DatabaseInventory: {
    DatabaseName: [
      {
        GlobalDatabaseName: "production.cubicrace.com",
        OracleSID: "production",
        Administrator: [
          {
            EmailAlias: "piyush",
            Extension: "6007",
            value: "Piyush Chordia"
          }
        ],
        DatabaseAttributes: {
          Type: "Production",
          Version: "9i"
        },
        Comments: "All new accounts need to be approved."
      },
      {
        GlobalDatabaseName: "development.cubicrace.com",
        OracleSID: "development",
        Administrator: [
          {
            EmailAlias: "kalpana",
            Extension: "6008",
            value: "Kalpana Pagariya"
          }
        ],
        DatabaseAttributes: {
          Type: "Development",
          Version: "9i"
        }
      }
    ]
  }
}
```

Source: <https://jsonexamples.wordpress.com/2018/03/23/difference-between-json-and-xml/>

The information in a statement is contained within a series of elements.

- **Version** – Specify the version of the policy language that you want to use. We recommend that you use the latest 2012-10-17 version. For more information, see [IAM JSON policy elements: Version](#)
- **Statement** – Use this main policy element as a container for the following elements. You can include more than one statement in a policy.
- **Sid** (Optional) – Include an optional statement ID to differentiate between your statements.
- **Effect** – Use Allow or Deny to indicate whether the policy allows or denies access.
- **Principal** (Required in some circumstances) – If you create a resource-based policy, you must indicate the account, user, role, or federated user to which you would like to allow or deny access. If you are creating an IAM permissions policy to attach to a user or role, you cannot include this element. The principal is implied as that user or role.
- **Action** – Include a list of actions that the policy allows or denies.
- **Resource** (Required in some circumstances) – If you create an IAM permissions policy, you must specify a list of resources to which the actions apply. If you create a resource-based policy, it depends on the resource you're using as to whether this element is required or not.
- **Condition** (Optional) – Specify the circumstances under which the policy grants permission.

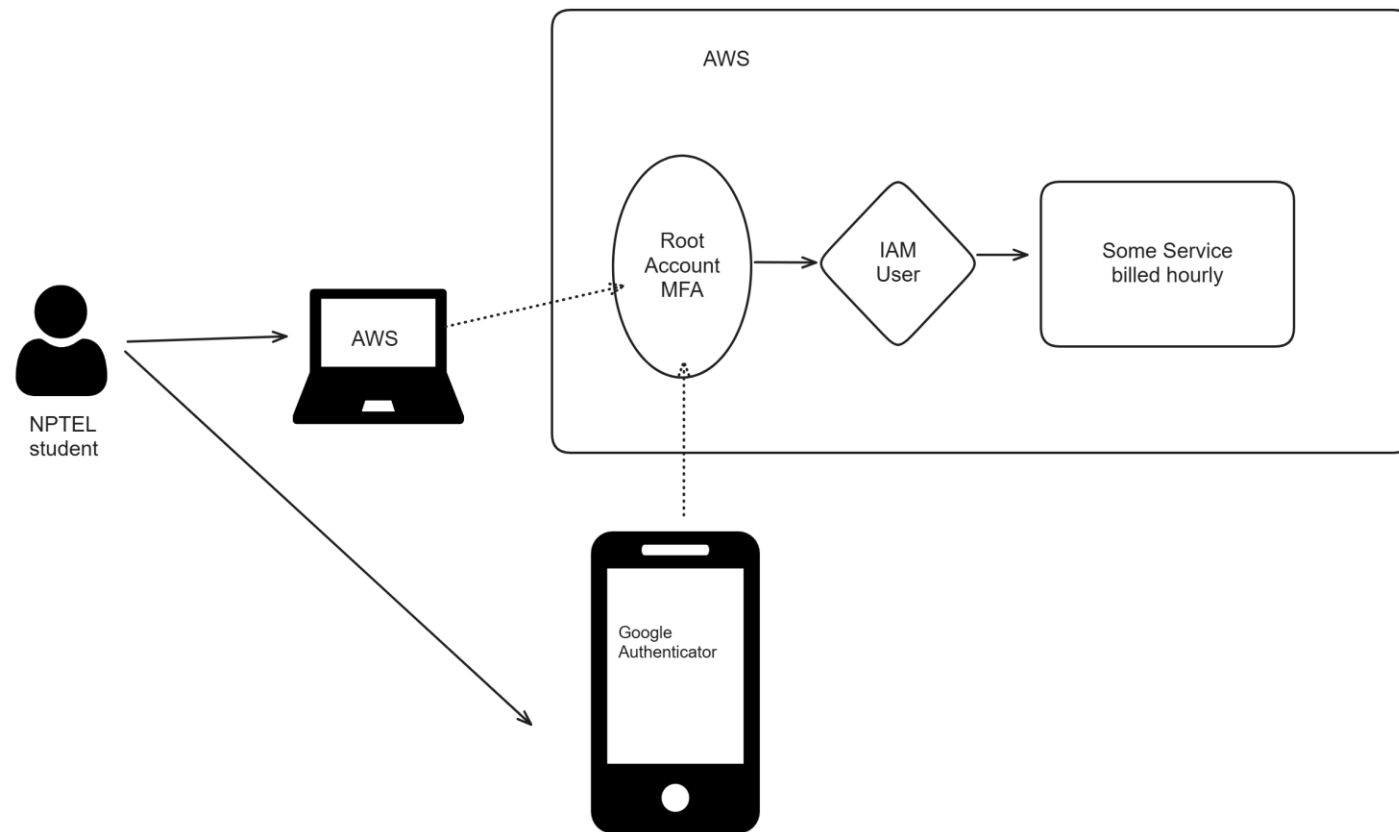
To learn about these and other more advanced policy elements, see [IAM JSON policy element reference](#).

# IAM policy

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "FirstStatement",
      "Effect": "Allow",
      "Action": ["iam:ChangePassword"],
      "Resource": "*"
    },
    {
      "Sid": "SecondStatement",
      "Effect": "Allow",
      "Action": "s3:ListAllMyBuckets",
      "Resource": "*"
    },
    {
      "Sid": "ThirdStatement",
      "Effect": "Allow",
      "Action": [
        "s3:List*",
        "s3:Get*"
      ],
      "Resource": [
        "arn:aws:s3:::amzn-s3-demo-bucket-confidential-data",
        "arn:aws:s3:::amzn-s3-demo-bucket-confidential-data/*"
      ],
      "Condition": {"Bool": {"aws:MultiFactorAuthPresent": "true"}}
    }
  ]
}
```



# MFA

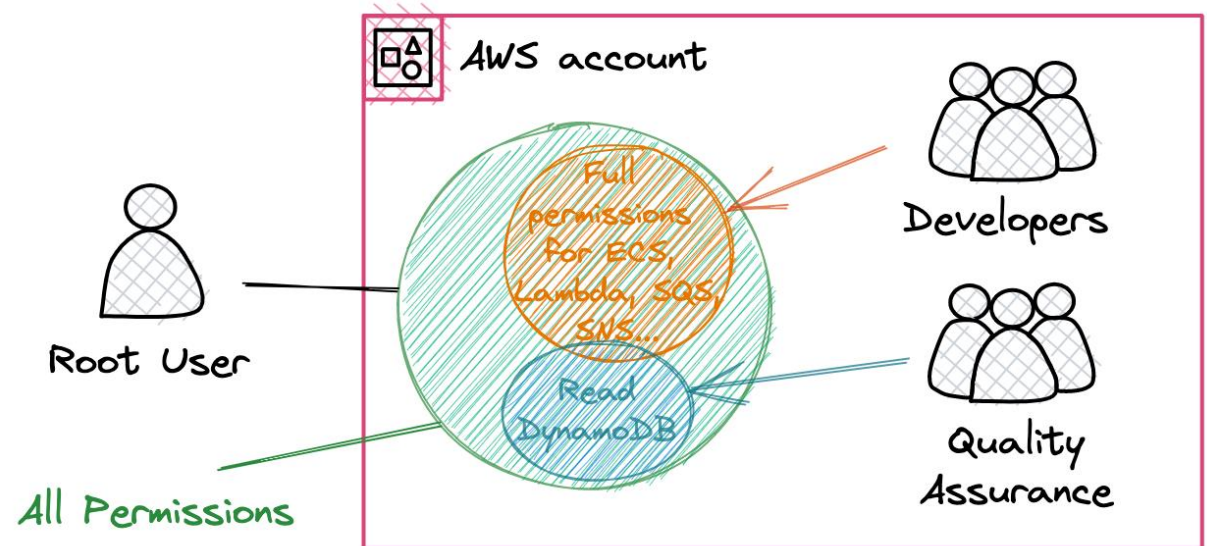
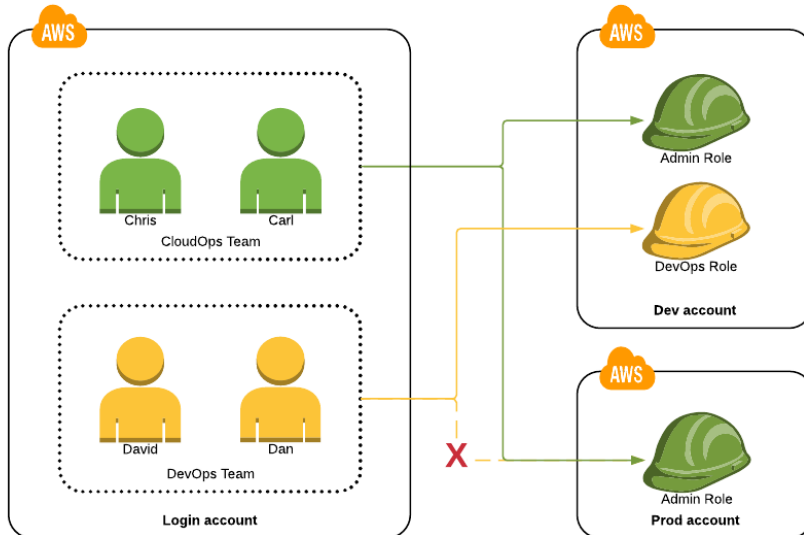
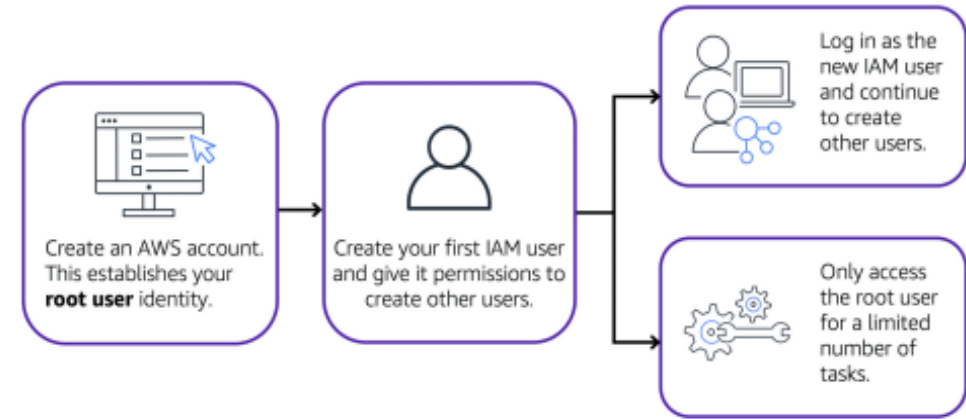


Source :<https://kevinkiruri.medium.com/enforcing-mfa-on-aws-66b228df699b>

# AWS Account

## Your Account's Root User :

- Usage of the root account, especially for daily tasks, introduces several major risks: even a small mistake when using the 'root' user can have catastrophic consequences due to the unrestricted permissions.
- AWS Identity and Access Management **IAM** user



# Checklist

## Step 1

- AWS Free-tier Account

## Step 2

- IAM User
- MULTI-FACTOR Authentication

## Step 3

- Budget
- zero spend + Monthly spend

# IAAS: Amazon EC2

An EC2 instance is a virtual server in the AWS Cloud. --> Close

Use Free-tier  
functions only

## Features of Amazon EC2

Amazon EC2 provides the following high-level features:

### Instances

Virtual servers.

### Amazon Machine Images (AMIs)

Preconfigured templates for your instances that package the components you need for your server (including the operating system and additional software).

### Instance types

Various configurations of CPU, memory, storage, networking capacity, and graphics hardware for your instances.

### Amazon EBS volumes

Persistent storage volumes for your data using Amazon Elastic Block Store (Amazon EBS).

### Instance store volumes

Storage volumes for temporary data that is deleted when you stop, hibernate, or terminate your instance.

### Key pairs

Secure login information for your instances. AWS stores the public key and you store the private key in a secure place.

### Security groups

A virtual firewall that allows you to specify the protocols, ports, and source IP ranges that can reach your instances, and the destination IP ranges to which your instances can connect

| Instance name ▾ | On-Demand hourly rate ▲ | vCPU ▾ | Memory ▾ | Storage ▾ | Network performance ▾ |
|-----------------|-------------------------|--------|----------|-----------|-----------------------|
| t4g.nano        | \$0.0042                | 2      | 0.5 GiB  | EBS Only  | Up to 5 Gigabit       |
| t3a.nano        | \$0.0047                | 2      | 0.5 GiB  | EBS Only  | Up to 5 Gigabit       |
| t3.nano         | \$0.0052                | 2      | 0.5 GiB  | EBS Only  | Up to 5 Gigabit       |
| t2.nano         | \$0.0058                | 1      | 0.5 GiB  | EBS Only  | Low                   |
| t4g.micro       | \$0.0084                | 2      | 1 GiB    | EBS Only  | Up to 5 Gigabit       |
| t3a.micro       | \$0.0094                | 2      | 1 GiB    | EBS Only  | Up to 5 Gigabit       |
| t3.micro        | \$0.0104                | 2      | 1 GiB    | EBS Only  | Up to 5 Gigabit       |
| t2.micro        | \$0.0116                | 1      | 1 GiB    | EBS Only  | Low to Moderate       |
| t4g.small       | \$0.0168                | 2      | 2 GiB    | EBS Only  | Up to 5 Gigabit       |

<https://aws.amazon.com/ec2/pricing/on-demand/>

<https://www.geeksforgeeks.org/amazon-ec2-creating-an-elastic-cloud-compute-instance/>

A blue oval with a dark blue border, containing the text "Use Free-tier functions only" in white.

Use Free-tier  
functions only

# Hands-on Amazon EC2

The End